MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12AX7WA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, miniature, high Mu

Outline	---	6-2 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	2g	2k	h	h	1a	1g	1k	hct

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rg/g	Ik/k	Pp/p	TE	Alt
Unit:	V	Vdc	Vdc	v	Meg	mAdc	W	°C	ft
Maximum:	13.8	330	0,-55	135	0.5	6.5	1.1	+165	(See note 1)
	6.9	---	---	---	---	---	---	---	---
Minimum:	11.4	---	---	---	---	---	---	---	---
	5.7	---	---	---	---	---	---	---	---
TEST CONDITIONS:	12.6	250	-2	0	---	---	---	---	---

GENERAL:

Qualification - Required.

Reliable tube

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFEC-TIVE	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.4	II	If	138	162	mA
1336	Heater-cathode leakage	See note 2	0.4	II	Ihk	---	20	µAdc
1266	Total grid current	Rg/g = 0.5 Meg (see notes 2 and 9)	0.4	II	Ic	0	-0.3	µAdc
1256	Electrode current (1) (anode)	See note 2	0.4	II	Ib	0.70	1.70	mAdc
1256	Electrode current (2) (anode)	Ec = -5 Vdc; Rp = 0.1 Meg (See note 2)	0.4	II	Ib	---	35	µAdc
1321	AC amplification	Ebb = 100 Vdc; Ecc = 0; Esig = 0.2 Vac; Rp = 0.5 Meg; Rg = 10 Meg; (see note 2)	0.4	II	Ep	8.4	---	Vac
1201	Short and discontin-uity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	See note 2	2.5	I	R	500	---	Meg
1256	Electrode current (1) (anode) (difference between sections)		2.5	I	Ib	---	0.6	mAdc
1306	Transconductance (1)	See note 2	2.5	I	Sm	1250	2050	µmhos
1306	Transconductance (2)	Ef = 11.4 V; (see note 2)	2.5	I	ΔSm Ef	---	15	%
1266	Grid emission	Ef = 15.0 V; Rg/g = 0.5 Meg; Ec = -12.0 Vdc (See notes 2 and 3)	2.5	I	Ic	0	-0.6	µAdc
2201	Noise and micro-phonics	Ef = 12.6 Vac; Ebb = 250 Vdc; Ecal = 85 mVac; Rp = 0.2 Meg; (see note 4)	2.5	I	---	---	---	---
1316	Amplification factor	See note 2	6.5	S3	Mu	85	115	---
1331	Direct-interelectrode capacitance	No shield (see note 2)			Cgp	1.30	2.10	pF
		No shield (see note 2)	6.5	Code E	Cin	1.20	1.00	pF
		No shield, section 1			Cout	0.22	0.70	pF
		No shield, section 2			Cout	0.10	0.58	pF

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFEC-TIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
1031	High-frequency vibration	Rp = 2,000 ohms; (see note 4)	6.5	Code E	Ep	---	150	mVac
1031	Sweep-frequency vibration	Rp = 2,000 ohms (see note 4)	6.5	Note 5	Ep	---	300	mVac
1041	Shock	630 G; Ehk = +100 Vdc (see note 6)	6.5	Note 5	---	---	---	---
1031	Vibration-fatigue		6.5	Note 5	---	---	---	---
---	Shock and vibration-fatigue test end points:							
1031	High-frequency vibration		---	---	Ep	---	225	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	30	µAdc
1321	AC amplification		---	---	Ep	7.4	----	Vac
1266	Total grid current		---	---	Ic	0	-0.5	µAdc
1121	Base strain	See note 7	---	---	---	---	---	---
2126	Envelope strain		2.5	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3					
1501	Heater cycling life	Ef = 7.5 V (heaters in parallel) Ebk = +135 Vdc; Ec = Eb = 0 1 min "on" 4 min "off"	---	---	---	---
---	Heater-cycling life-test end point					
1336	Heater cathode leakage		Ibk	---	30	µAdc
1516	Stability life	Eb = 300 Vdc; Ec = -1.6 Vdc; Ebk = +135 Vdc; Rg/g = 0.5 Meg; TA = room	---	---	---	---
	Stability life-test end point (2 and 20 hours)					---
1321	Change in AC amplification of individual tubes		ΔRp t	---	10	%
1501	Intermittent life	Eb = 300 Vdc; Ec = -1.6 Vdc; Ebk = +135 Vdc; Rg/g = 0.5 Meg; TA = room TE = +165°C (min) (see note 8)group E	---	---	---	---
---	Life-test end points (intermittent) (1,000 hours):					
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic	0	-0.3	µAdc
1301	Heater current		If	138	166	mA
1321	Change in AC amplification of individual tubes		ΔRp t	---	20	%
1336	Heater-cathode leakage		Ibk	---	20	µAdc
1211	Insulation of electrodes		R	200	---	MΩ

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NOTES:

1. See "Reduced pressure (altitude) rating," and altitude, maximum peak voltage.

2. Test each unit separately.

3. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions specified below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ec	Eb	Rk/k	Rg/g
V	Vdc	Vdc	Ohms	Meg
15.0	-1.5	300	0	0.5

4. Tie 1k to 2k; 1g to 2g; and 1a to 2a.

5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.

6. A grid resistor of 0.1 megohm shall be added to each section; however, this resistor shall not be used when a thyratron-type short indicator is employed.

7. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

   a. Accepted if not more than one defective for class "A", "B", or "C" defects, respectively (see method 1121, or if not more than a total of two defectives are found in the sample.

   b. Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (+5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. This test shall be performed at the conclusion of the holding period.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Navy - EC
Air Force - 85

Review activities:
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User activities:
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Preparing activity:
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